

EDUCATION	PhD in Physics at CUNY Graduate Center, New York, NY	2016–2022
	Dissertation: <i>Collapse of Two-Dimensional Magnetic Skyrmions</i>	
	B.S. in Physics and Mechanical Engineering at Tufts University	2011–2015
MANUSCRIPTS IN PREP	Derras-Chouk, Amel, Zhengzhao Johnny Luo, and Gregory Elsaesser (In prep). <i>A Satellite-based Estimate of Deep Convective Precipitation Efficiency in the Tropics</i> .	
	Derras-Chouk, Amel et al. (Submitted). <i>Estimating Vertical Velocity from Temperature, Pressure, and Latent Heating</i> .	
PUBLICATIONS	Braneon, Christian et al. (2024). “NPCC4: New York City climate risk information 2022—observations and projections”. In: <i>Annals of the New York Academy of Sciences</i> 1539.1, pp. 13–48.	
	Derras-Chouk, Amel and Zhengzhao Johnny Luo (2024). “A geostationary satellite-based approach to estimate convective mass flux and revisit the hot tower hypothesis”. In: <i>Surveys in Geophysics</i> 45.6, pp. 1959–1977.	
	Derras-Chouk, Amel, Eugene M Chudnovsky, and Dmitry A Garanin (2022). “Dynamics of the collapse of a ferromagnetic skyrmion in a centrosymmetric lattice”. In: <i>Physical Review B</i> 105.13, p. 134432.	
	Derras-Chouk, Amel and Eugene M Chudnovsky (2021). “Skyrmions near defects”. In: <i>Journal of Physics: Condensed Matter</i> 33.19, p. 195802.	
	Derras-Chouk, Amel, Eugene M Chudnovsky, and Dmitry A Garanin (2021). “Quantum states of a skyrmion in a two-dimensional antiferromagnet”. In: <i>Physical Review B</i> 103.22, p. 224423.	
	– (2019). “Thermal collapse of a skyrmion”. In: <i>Journal of Applied Physics</i> 126.8.	
	– (2018). “Quantum collapse of a magnetic skyrmion”. In: <i>Physical Review B</i> 98.2, p. 024423.	
	Derras-Chouk, Amel, Eugene M Chudnovsky, Dmitry A Garanin, and Reem Jaafar (2018). “Graphene cantilever under Casimir force”. In: <i>Journal of Physics D: Applied Physics</i> 51.19, p. 195301.	
RESEARCH EXPERIENCE	Postdoctoral Research Fellow , Columbia Climate School	Sept. 2024–present
	Postdoctoral Researcher , The City College of New York, CUNY	Sept. 2022–Aug. 2024
	Graduate Student Intern , NASA Goddard Institute for Space Studies	Oct. 2021–Aug. 2022
	Graduate Student Researcher , CUNY Graduate Center	July 2017–Sept. 2022
TEACHING EXPERIENCE	Ajunct Lecturer Lehman College	Aug. 2017–May 2021
	– Lab Section of General Physics 1 (PHY 166/168)	
	– Lab Section of General Physics 2 (PHY 167/169)	

Computer Science Teacher SA Charter Schools

Sept. 2015–May 2016

– Middle school coding elective

Teaching Assistant Tufts University

Sept. 2014–May 2015

– Lab Section of Electronics (PHY 0041)

STOMP Fellow, CEEO at Tufts University

Sept. 2014–May 2015

Through Tufts’s Center for Engineering Education and Outreach (CEEEO) Student Teacher Outreach Mentorship Program (STOMP), I taught engineering concepts to students in local elementary schools using LEGO robotics kits.

SELECTED
PRESENTATIONS

- Sept. 2025 Invited seminar at NOAA Geophysical Fluid Dynamics Laboratory, Princeton, NJ, USA. *Quantifying Vertical Motion Using Satellite Data.*
- July 2025 Gordon Research Conference on Atmospheric Radiation (Poster presentation), Lewiston, ME, USA. *Estimating vertical velocity from temperature, pressure, and latent heating.*
- Apr. 2025 NASA Goddard Institute for Space Science, New York, NY, USA. *Quantifying Convective Vertical Velocity and Precipitation Efficiency Using Satellite Data.*
- Feb. 2025 CloudSat Calipso Science Team Meeting, Fort Collins, CO, USA. *Estimating convective mass flux, detrainment, and precipitation efficiency using CloudSat, Geostationary, and IMERG data.*
- Dec. 2023 AGU Annual meeting (Poster presentation), San Francisco, CA. *A Satellite-Based Approach to Estimating Convective Mass Flux and Revisiting the Hot Tower Hypothesis*

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Sept. 2017–Dec. 2023

HONORS AND
AWARDS

CUNY Graduate Center Dissertation Fellowship

2020–2021

Physics Tithe Fellowship

2018–2022